# R85/B_gradient_balance | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 26 | update 37440 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) |
|---|---|---|---|---|---|---|---|---|
| VAL ALL | 70.9358 | 66.2037 | 11.416845 | 11.427978 | 11.371692 | -0.1910 | +0.3967 | 0.4059 |
| TEST ALL | 71.2111 | 66.1802 | 11.373750 | 11.385299 | 11.329932 | -0.1963 | +0.3869 | 0.3949 |

| Dataset | n | NSS Top1 (%) | R85 Top1 (%) | NSS cost | R85 cost |
|---|---|---|---|---|---|
| TSP TEST | 4500 | 45.888889 | 46.844444 | 7.970400719 | 7.971097270 |
| CVRP TEST | 4500 | 45.977778 | 47.355556 | 18.004802368 | 18.007877977 |
| TSPLIB | 49 | 20.408163 | 24.489796 | 8.217907074 | 8.208315315 |
| CVRPLIB | 100 | 70.000000 | 61.000000 | 67.101111717 | 67.246580114 |

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: OVR BCE on raw utility

# R85/B_gradient_balance | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) | TRAIN SBS |
|---|---|---|---|---|---|---|---|---|---|
| ALL | 70.9358 | 66.2037 | 11.416845 | 11.427978 | 11.371692 | -0.1910 | +0.3967 | 0.4059 | per-problem TRAIN SBS |
| TSP | 46.7111 | 33.1111 | 8.024859 | 8.045354 | 7.990533 | -0.2547 | +0.4296 | 0.4175 | T2T |
| CVRP | 45.7778 | 18.7111 | 18.157119 | 18.269377 | 18.064087 | -0.6145 | +0.5150 | 0.6771 | RELD_CVRP |
| ATSP | 76.7556 | 47.4889 | 1.638979 | 1.676905 | 1.632471 | -2.2616 | +0.3987 | 0.3945 | UNICO_MatPOENet |
| OVRP | 40.4000 | 35.7556 | 7.462928 | 7.469579 | 7.415777 | -0.0890 | +0.6358 | 0.6505 | MoSES_RF |
| VRPB | 81.0667 | 80.8444 | 9.353692 | 9.354876 | 9.318750 | -0.0127 | +0.3750 | 0.3846 | reld_mt |
| VRPL | 38.4000 | 33.0889 | 12.431619 | 12.437959 | 12.383444 | -0.0510 | +0.3890 | 0.3962 | MoSES_RF |
| VRPTW | 79.9111 | 79.9556 | 18.819224 | 18.819431 | 18.766139 | -0.0011 | +0.2829 | 0.2993 | reld_mt |
| OVRPTW | 83.8444 | 83.7778 | 10.143288 | 10.143730 | 10.116859 | -0.0044 | +0.2612 | 0.2581 | reld_mt |
| OVRPB | 88.5111 | 88.5111 | 6.357217 | 6.357254 | 6.341078 | -0.0006 | +0.2545 | 0.2644 | reld_mt |
| OVRPL | 40.2889 | 35.9111 | 7.439946 | 7.445709 | 7.390416 | -0.0774 | +0.6702 | 0.6870 | MoSES_RF |
| VRPBL | 78.9111 | 78.5111 | 9.368195 | 9.371523 | 9.328366 | -0.0355 | +0.4270 | 0.4290 | reld_mt |
| VRPBTW | 78.4000 | 78.4222 | 19.948652 | 19.950430 | 19.851084 | -0.0089 | +0.4915 | 0.4702 | reld_mt |
| VRPLTW | 78.8444 | 78.7556 | 18.764513 | 18.765871 | 18.707843 | -0.0072 | +0.3029 | 0.3188 | reld_mt |
| OVRPBL | 89.2667 | 89.2000 | 6.369700 | 6.370093 | 6.352824 | -0.0062 | +0.2657 | 0.2769 | reld_mt |
| OVRPBTW | 83.5111 | 83.4444 | 10.660751 | 10.661098 | 10.623245 | -0.0033 | +0.3531 | 0.3259 | reld_mt |
| OVRPLTW | 83.5778 | 83.4889 | 10.157191 | 10.157423 | 10.129818 | -0.0023 | +0.2702 | 0.2738 | reld_mt |
| VRPBLTW | 79.4444 | 79.4000 | 19.795096 | 19.797044 | 19.706420 | -0.0098 | +0.4500 | 0.4364 | reld_mt |
| OVRPBLTW | 83.2222 | 83.2889 | 10.610239 | 10.609955 | 10.571302 | +0.0027 | +0.3683 | 0.3457 | reld_mt |

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

# R85/B_gradient_balance | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) | TRAIN SBS |
|---|---|---|---|---|---|---|---|---|---|
| ALL | 71.2111 | 66.1802 | 11.373750 | 11.385299 | 11.329932 | -0.1963 | +0.3869 | 0.3949 | per-problem TRAIN SBS |
| TSP | 46.8444 | 34.5333 | 7.971097 | 7.989621 | 7.937593 | -0.2318 | +0.4221 | 0.4153 | T2T |
| CVRP | 47.3556 | 17.9333 | 18.007878 | 18.124983 | 17.918509 | -0.6461 | +0.4988 | 0.6484 | RELD_CVRP |
| ATSP | 76.5333 | 47.2444 | 1.635563 | 1.674207 | 1.629441 | -2.3082 | +0.3757 | 0.3736 | UNICO_MatPOENet |
| OVRP | 40.4444 | 33.6444 | 7.446547 | 7.455750 | 7.400487 | -0.1234 | +0.6224 | 0.6339 | MoSES_RF |
| VRPB | 80.5778 | 80.4000 | 9.350813 | 9.351810 | 9.315298 | -0.0107 | +0.3813 | 0.3834 | reld_mt |
| VRPL | 37.4444 | 32.2667 | 12.350620 | 12.356138 | 12.300958 | -0.0447 | +0.4037 | 0.4110 | MoSES_RF |
| VRPTW | 81.1111 | 80.8444 | 18.689392 | 18.690545 | 18.637408 | -0.0062 | +0.2789 | 0.2913 | reld_mt |
| OVRPTW | 83.9333 | 83.9333 | 10.092782 | 10.093063 | 10.068135 | -0.0028 | +0.2448 | 0.2363 | reld_mt |
| OVRPB | 88.0889 | 88.1333 | 6.371030 | 6.370826 | 6.352225 | +0.0032 | +0.2960 | 0.3037 | reld_mt |
| OVRPL | 41.1111 | 34.6000 | 7.464807 | 7.472542 | 7.418033 | -0.1035 | +0.6305 | 0.6429 | MoSES_RF |
| VRPBL | 79.6222 | 79.4222 | 9.395116 | 9.396379 | 9.355967 | -0.0134 | +0.4184 | 0.4211 | reld_mt |
| VRPBTW | 78.7778 | 78.7556 | 19.664508 | 19.665725 | 19.577666 | -0.0062 | +0.4436 | 0.4393 | reld_mt |
| VRPLTW | 79.3111 | 79.2444 | 18.801905 | 18.803136 | 18.745143 | -0.0065 | +0.3028 | 0.3192 | reld_mt |
| OVRPBL | 89.2667 | 89.2889 | 6.375684 | 6.375741 | 6.359310 | -0.0009 | +0.2575 | 0.2688 | reld_mt |
| OVRPBTW | 84.9111 | 84.8222 | 10.658895 | 10.659591 | 10.624748 | -0.0065 | +0.3214 | 0.2941 | reld_mt |
| OVRPLTW | 82.9111 | 82.8667 | 10.085869 | 10.086112 | 10.059001 | -0.0024 | +0.2671 | 0.2677 | reld_mt |
| VRPBLTW | 79.9111 | 79.7333 | 19.703247 | 19.707068 | 19.613284 | -0.0194 | +0.4587 | 0.4405 | reld_mt |
| OVRPBLTW | 83.6444 | 83.5778 | 10.661749 | 10.662149 | 10.625574 | -0.0037 | +0.3405 | 0.3173 | reld_mt |

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

# R85/B_gradient_balance | LIB and locked provenance

Current R85 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) |
|---|---|---|---|---|---|---|---|---|
| TSPLIB | 24.4898 | 10.2041 | 8.208315 | 8.240435 | 8.142167 | -0.3898 | +0.8124 | 0.8770 |
| CVRPLIB | 61.0000 | 45.0000 | 67.246580 | 67.245364 | 66.945093 | +0.0018 | +0.4503 | 0.5444 |

SHA256: 5361f35200ded36a1ed5d5e847f2d9537d03b3d0fc36c9943496fa396f84f870

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, original R77 maximin logits, stable ascending-global-solver-ID tie break

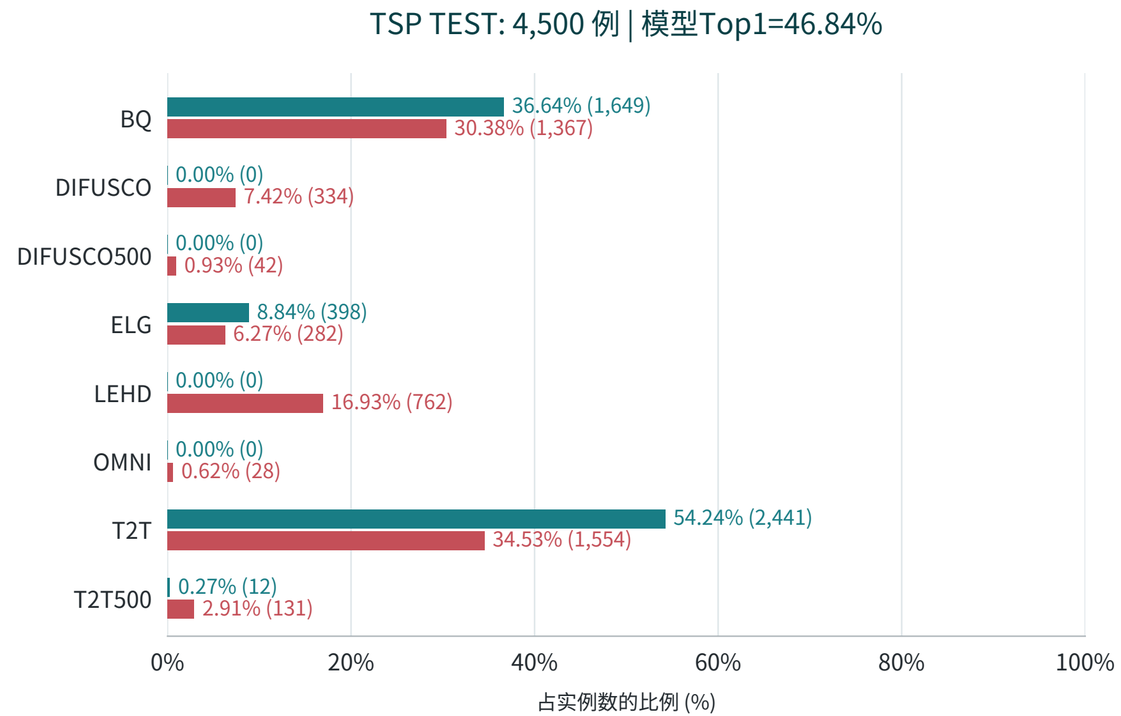
Normalization: new81000 TRAIN only; no TEST/LIB refit.

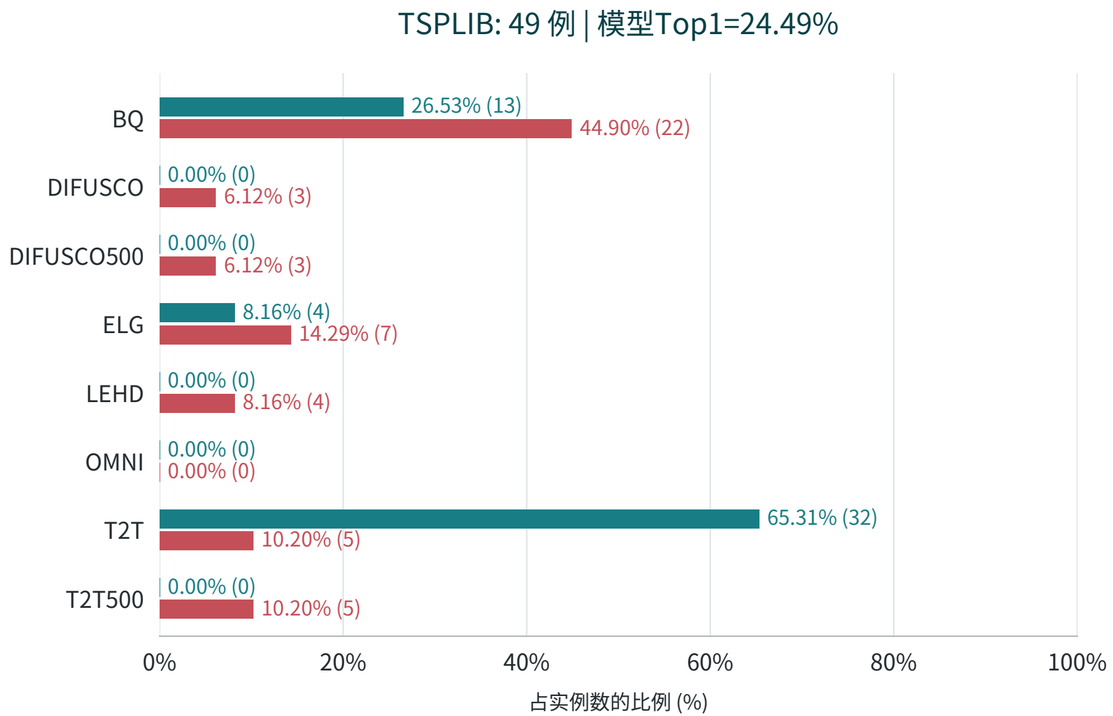
# R85/B_gradient_balance | TSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

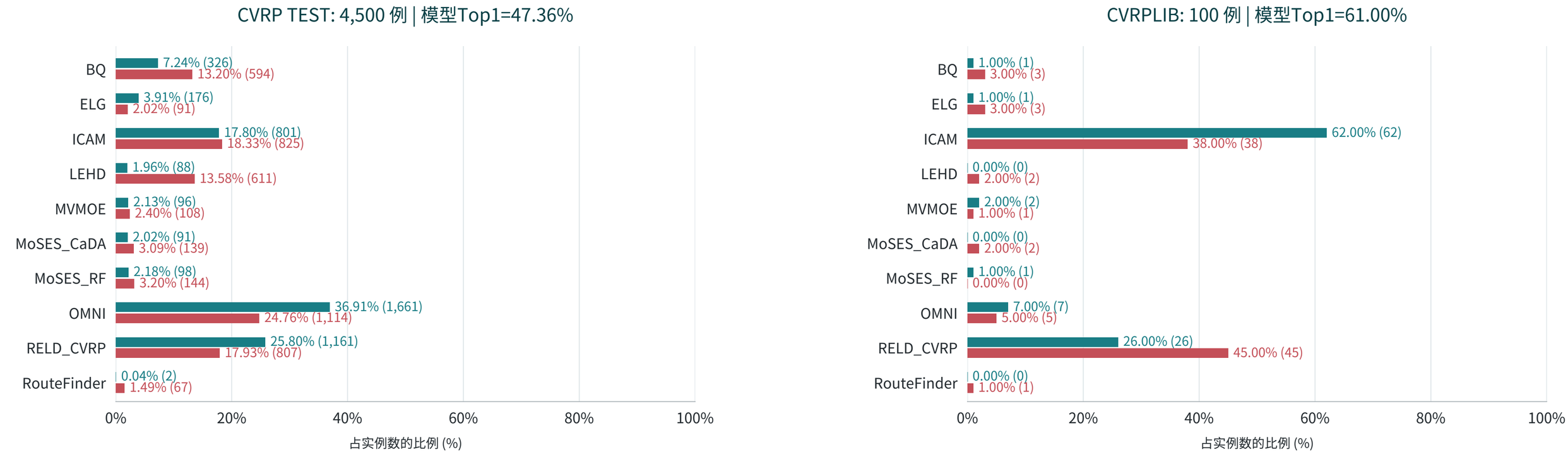
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | CVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

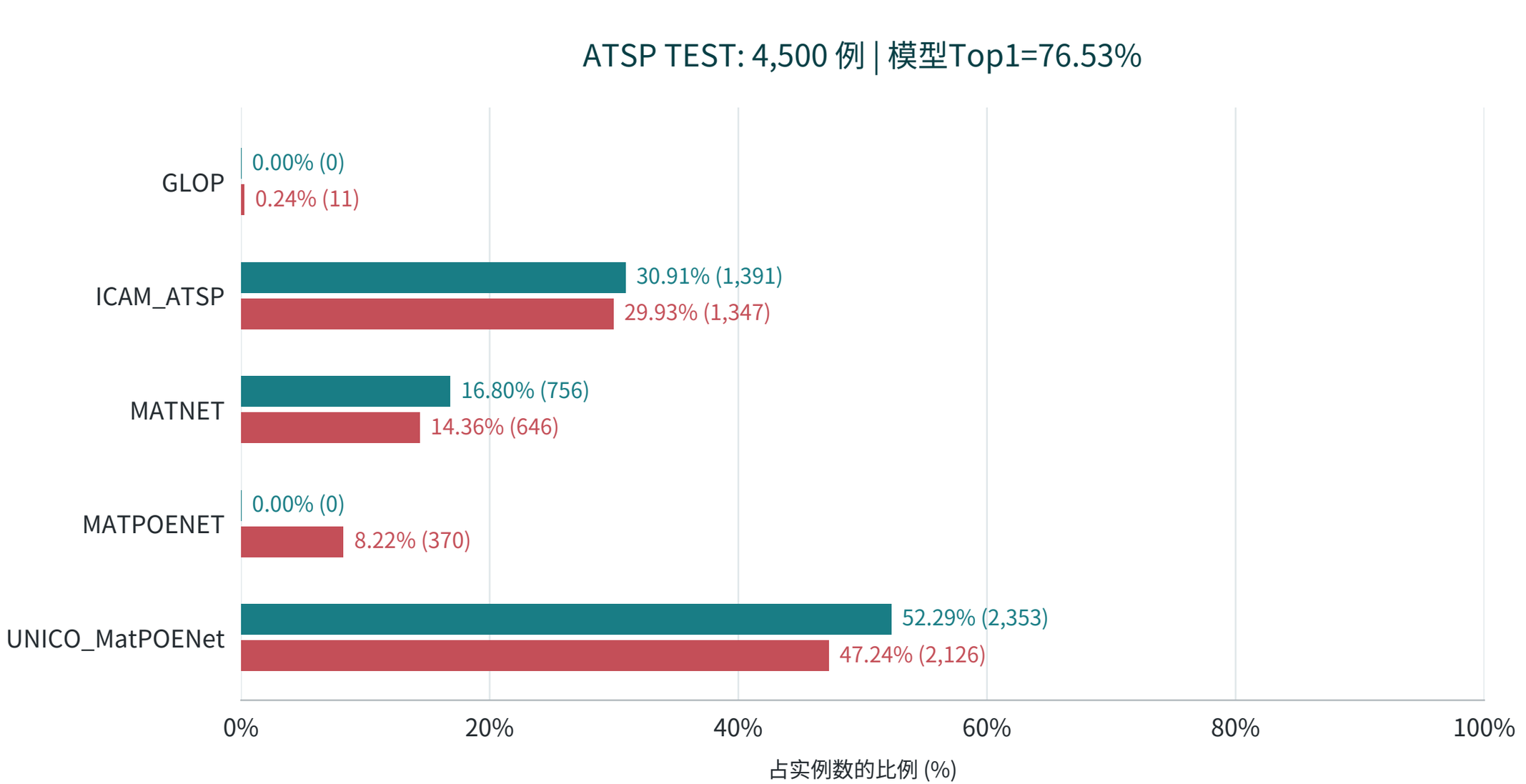
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | ATSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

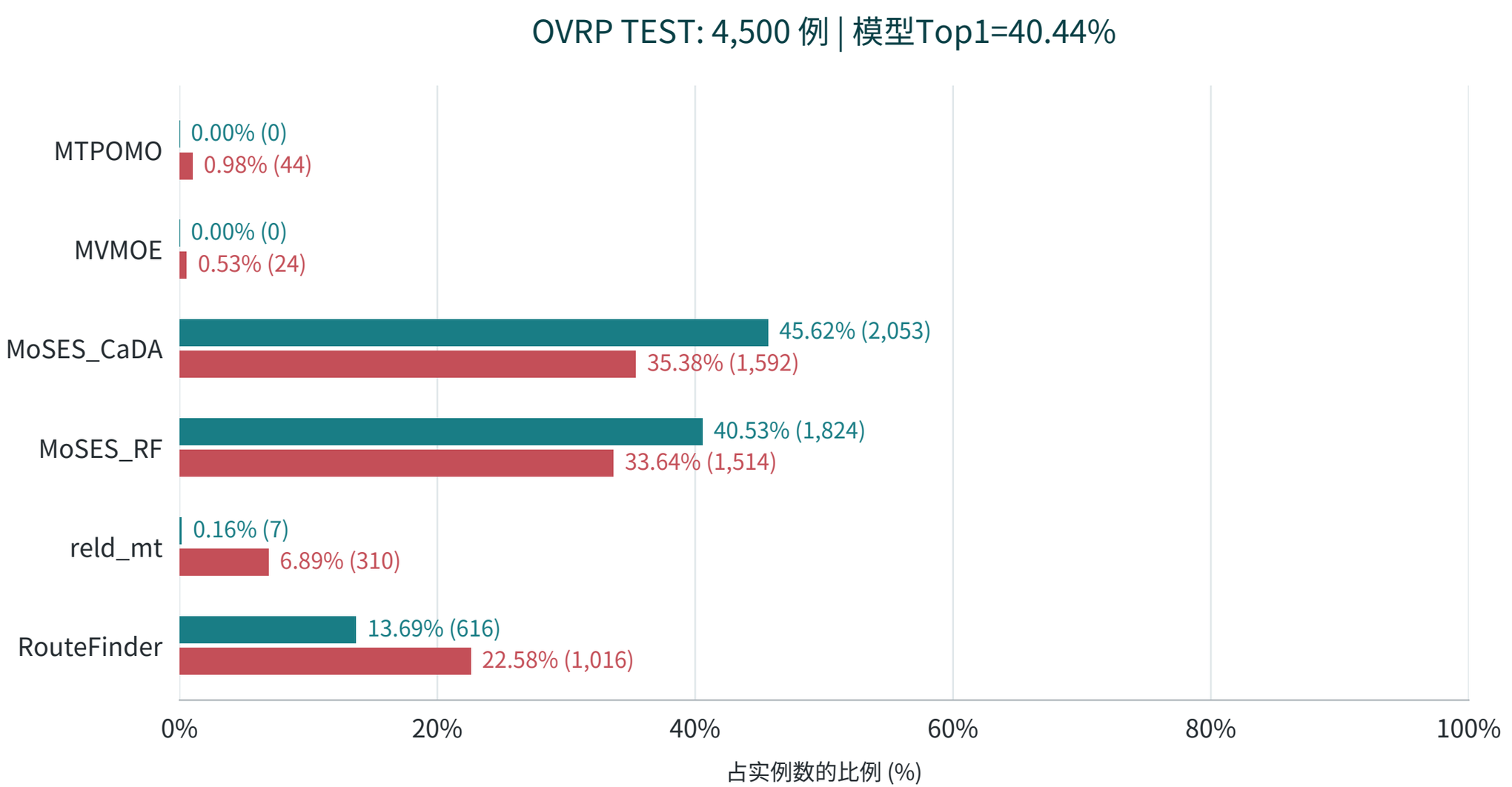
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

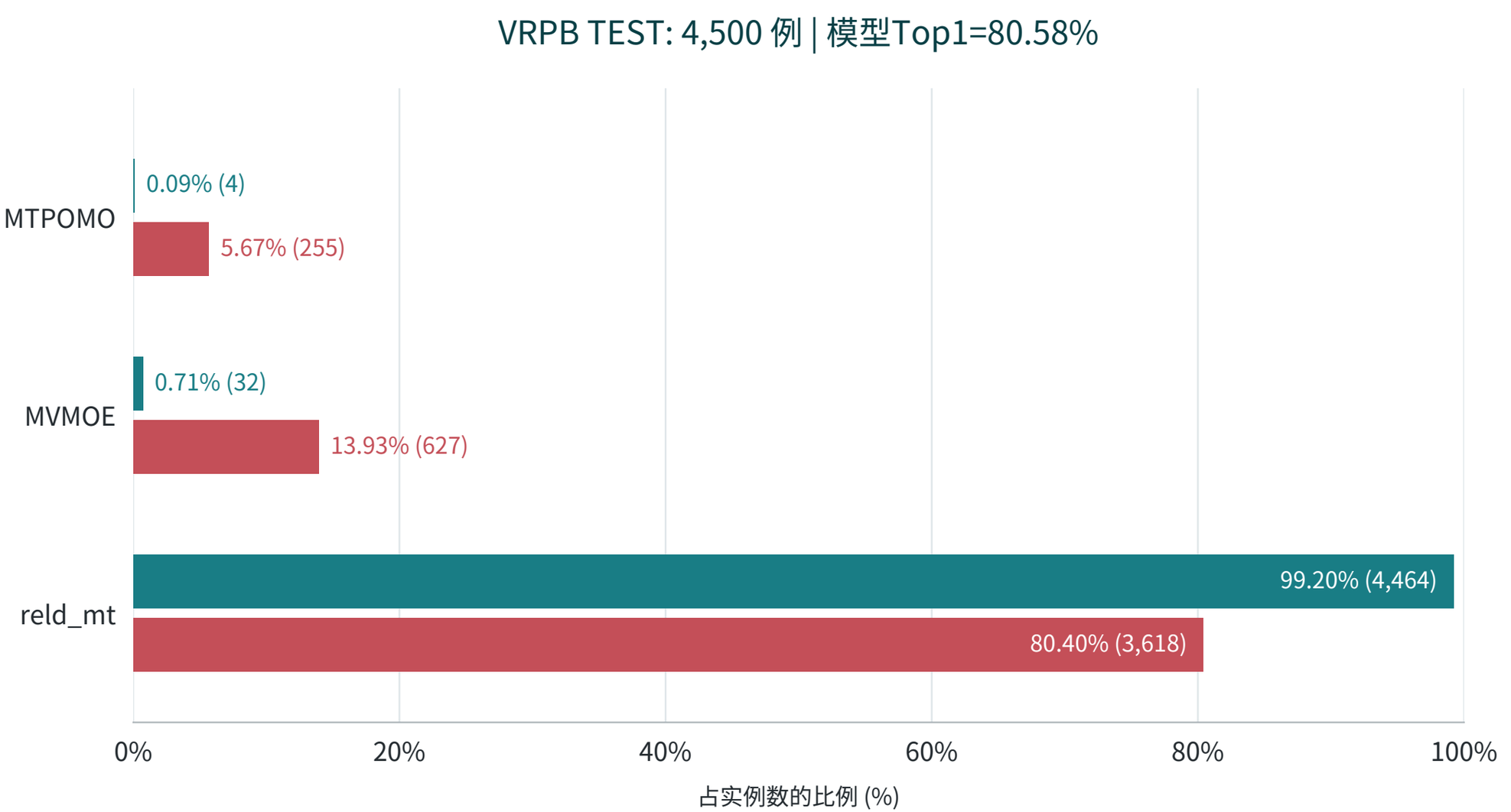
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

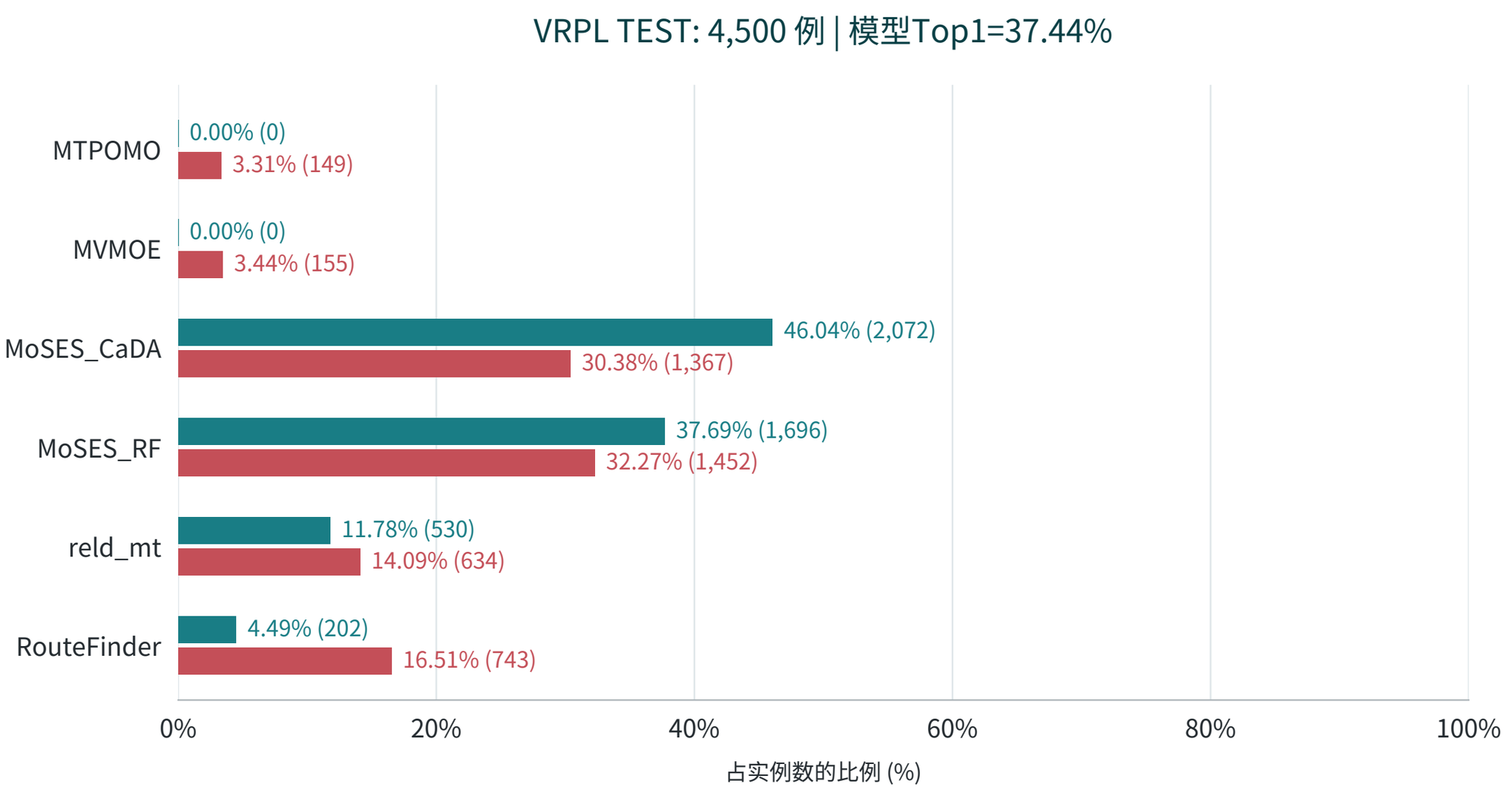
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

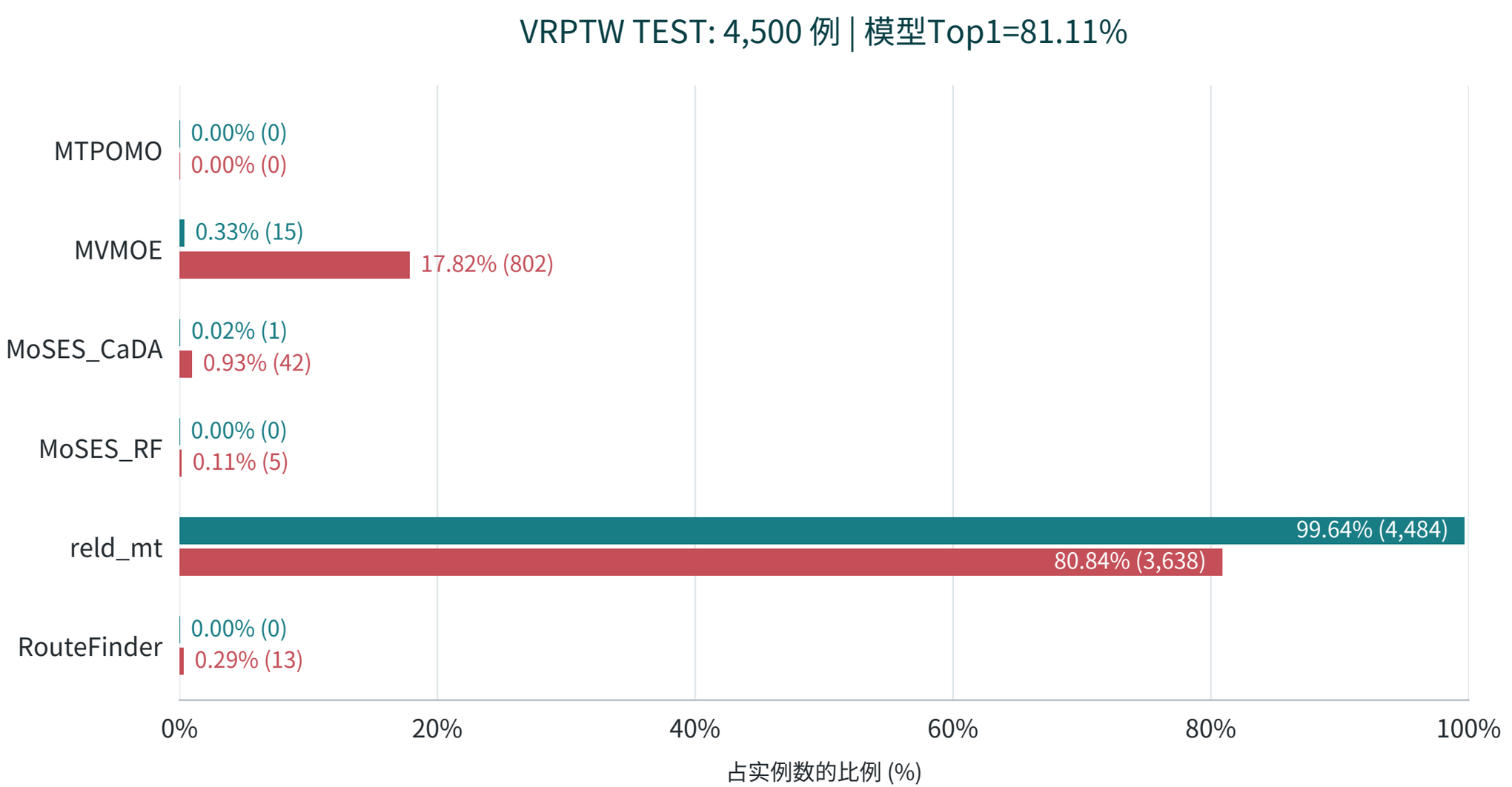
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

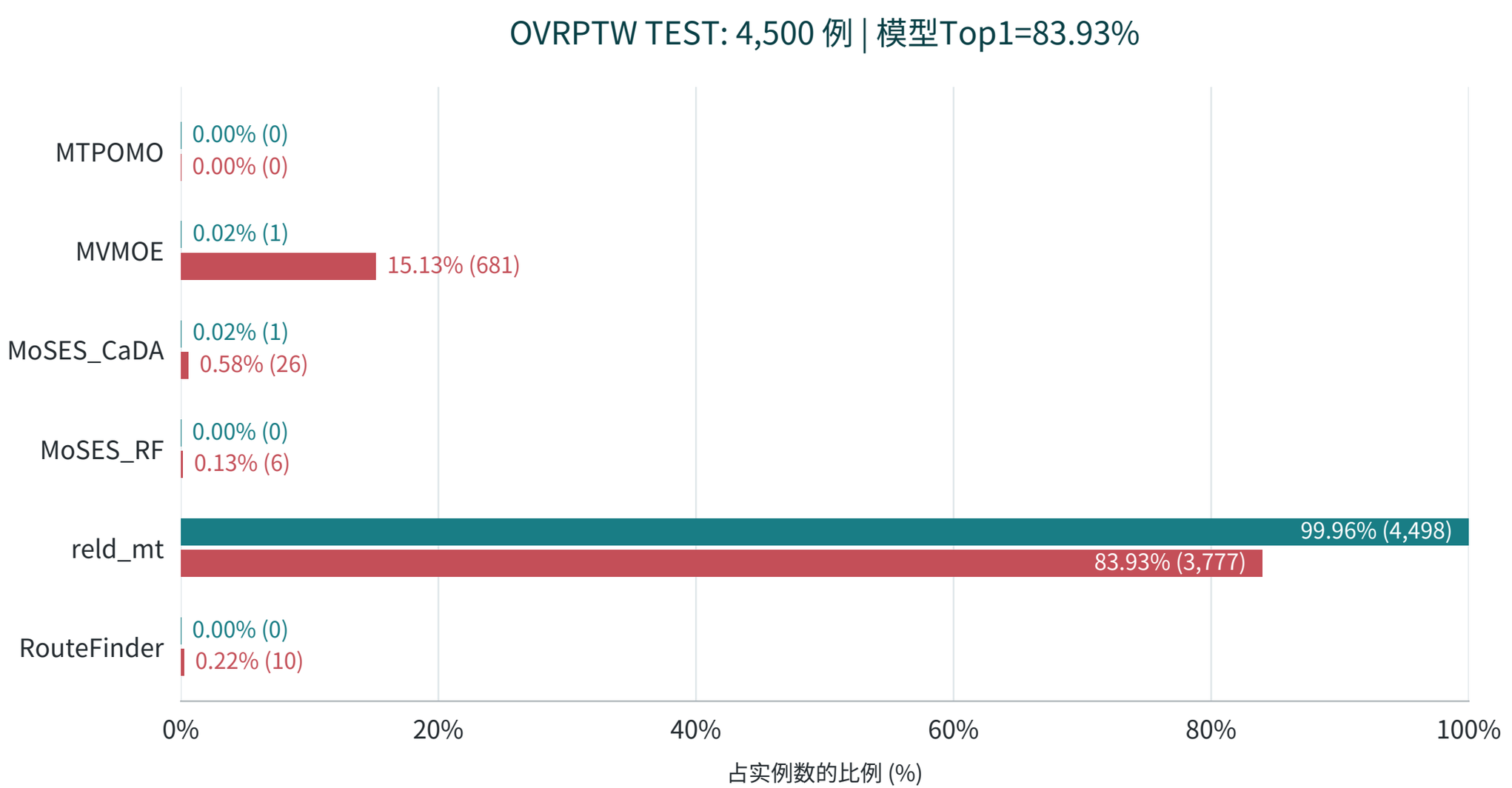
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

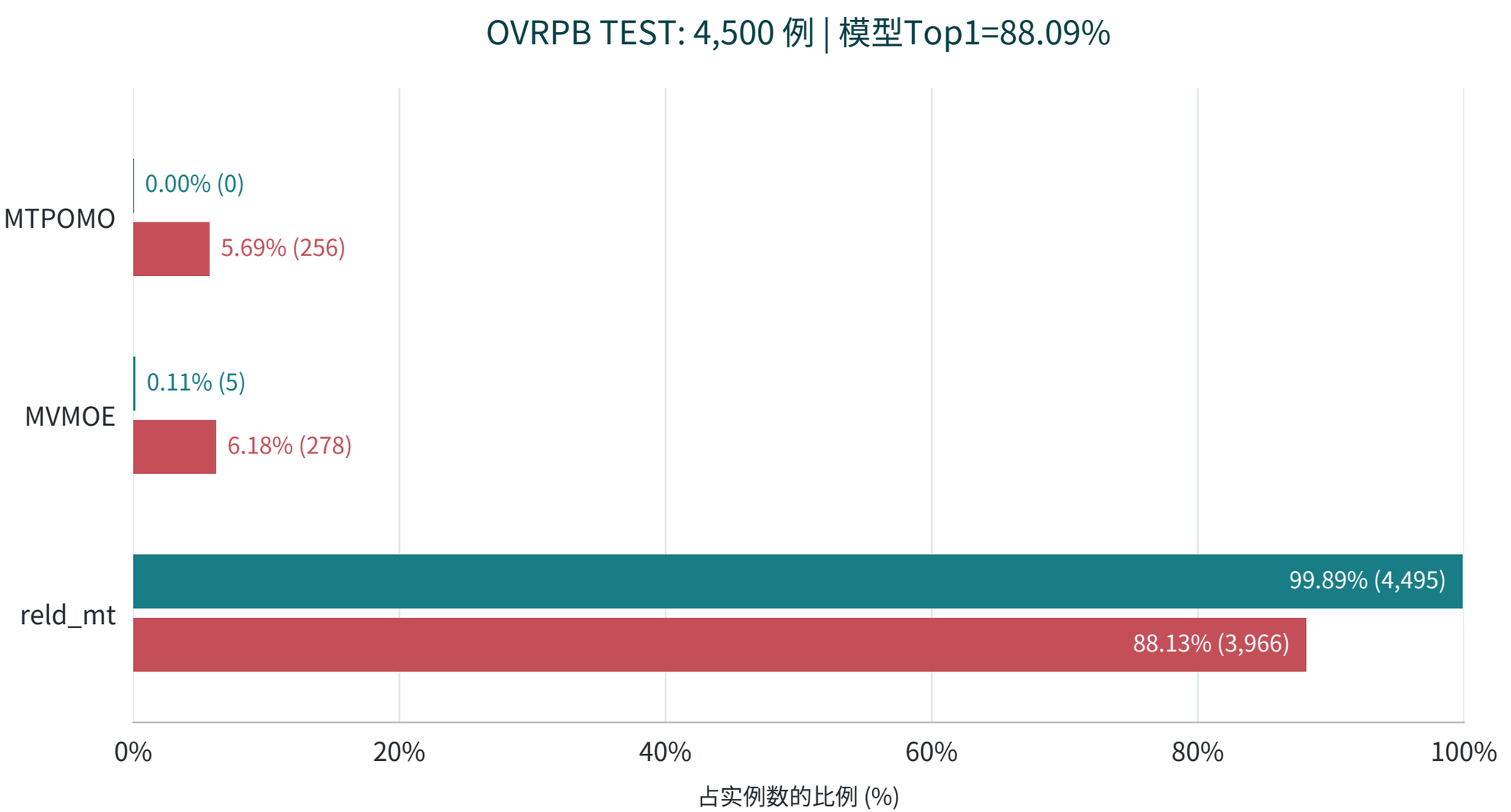
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

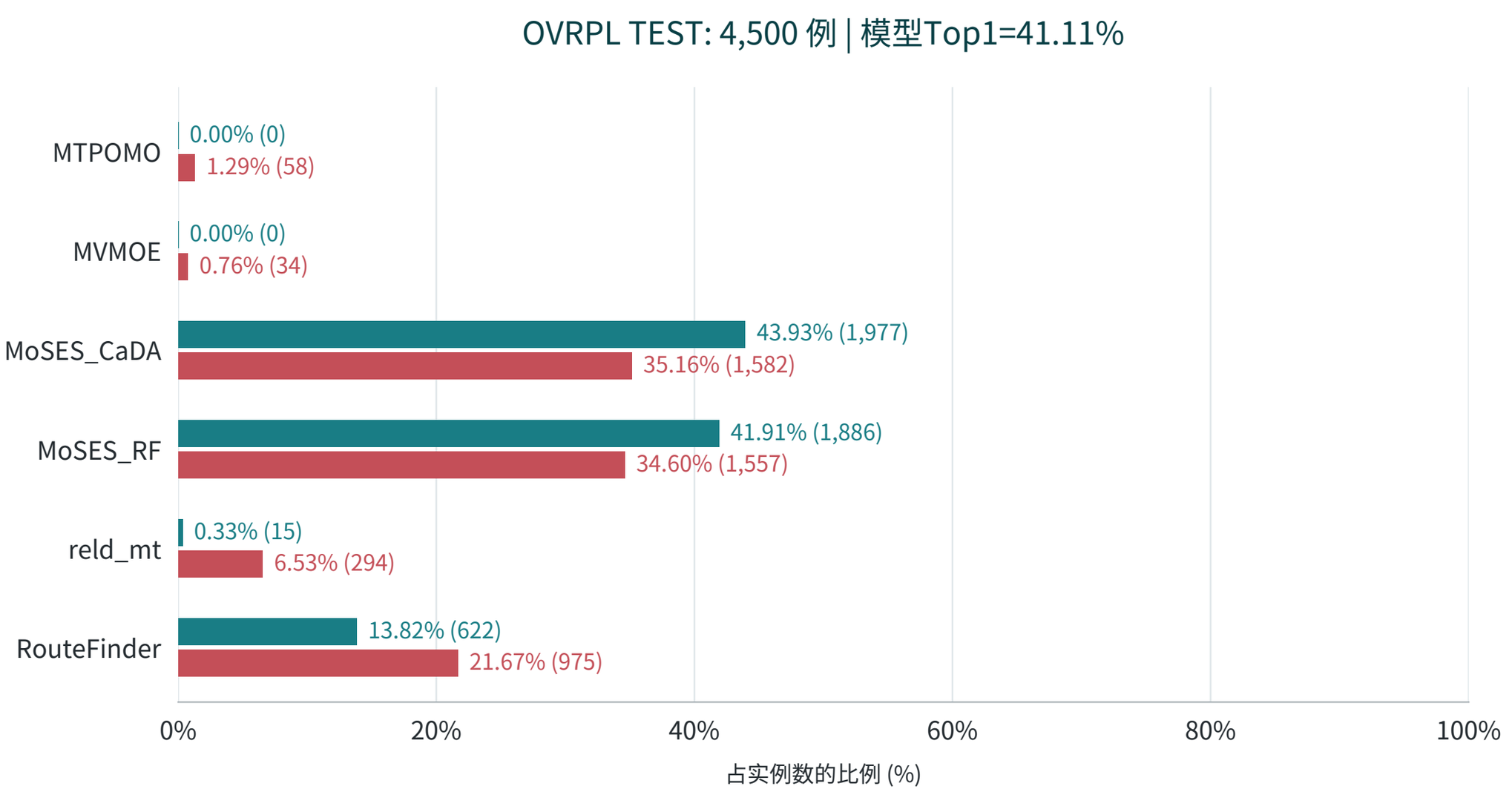
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

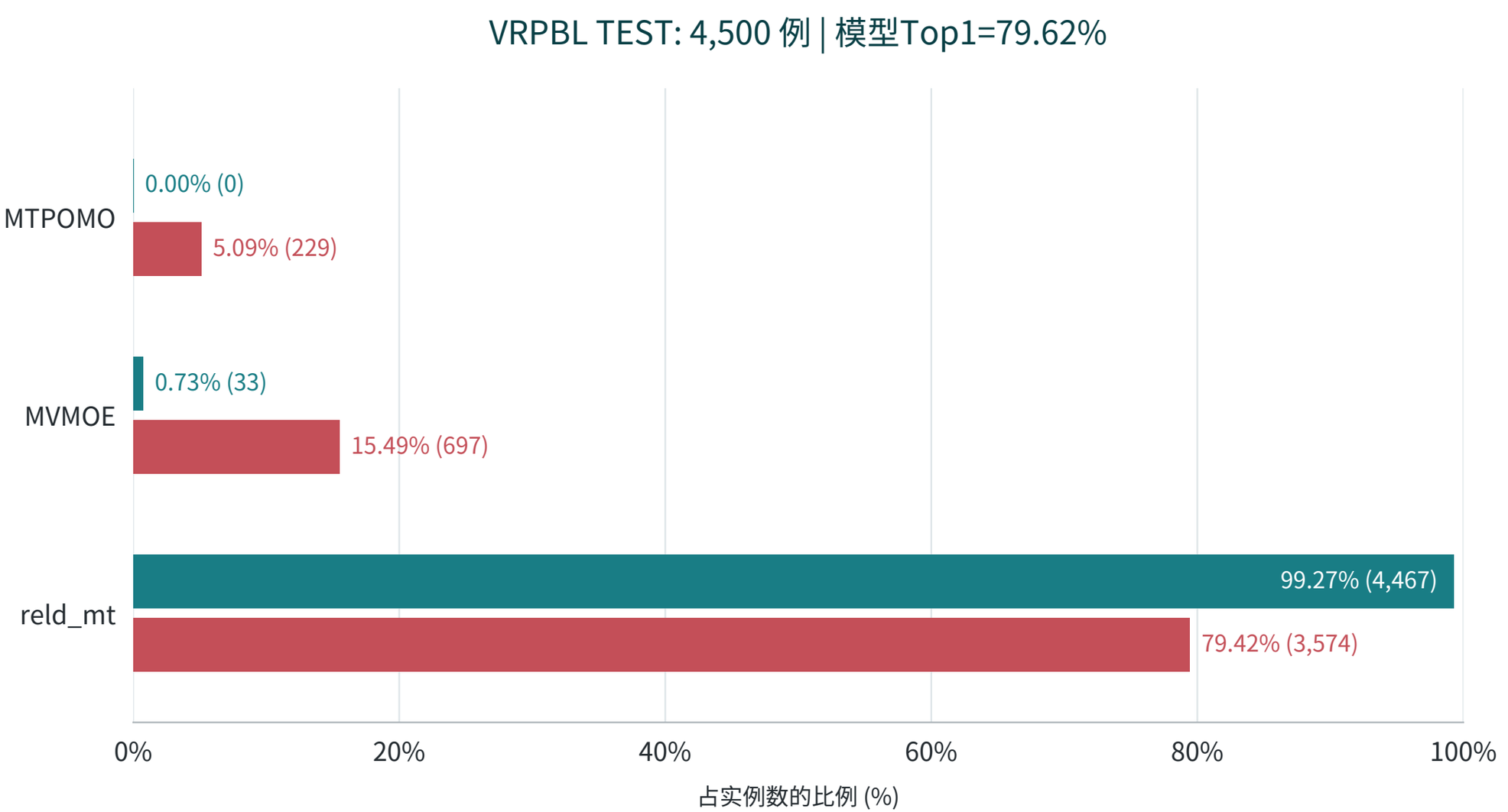
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

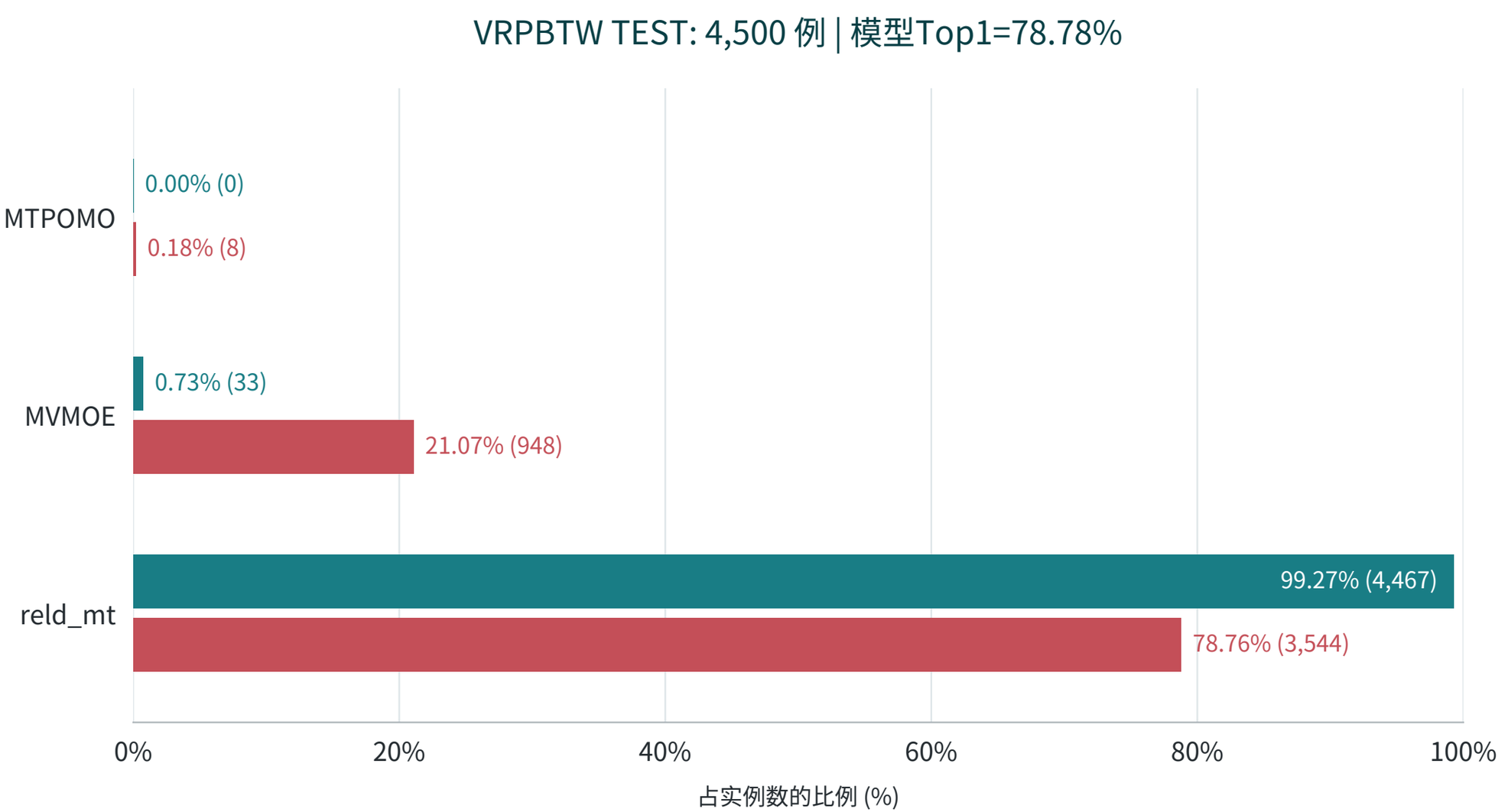
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

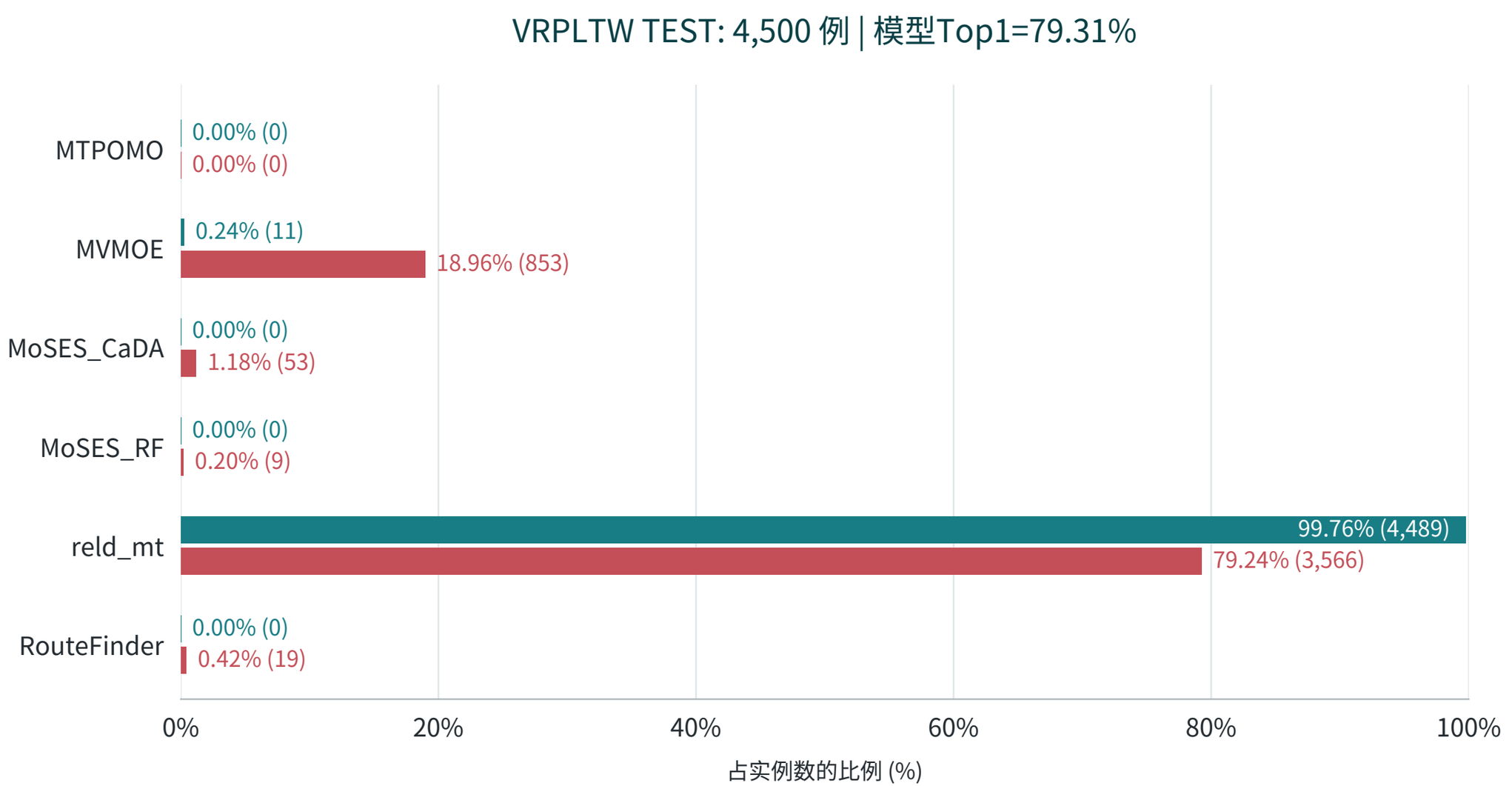
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

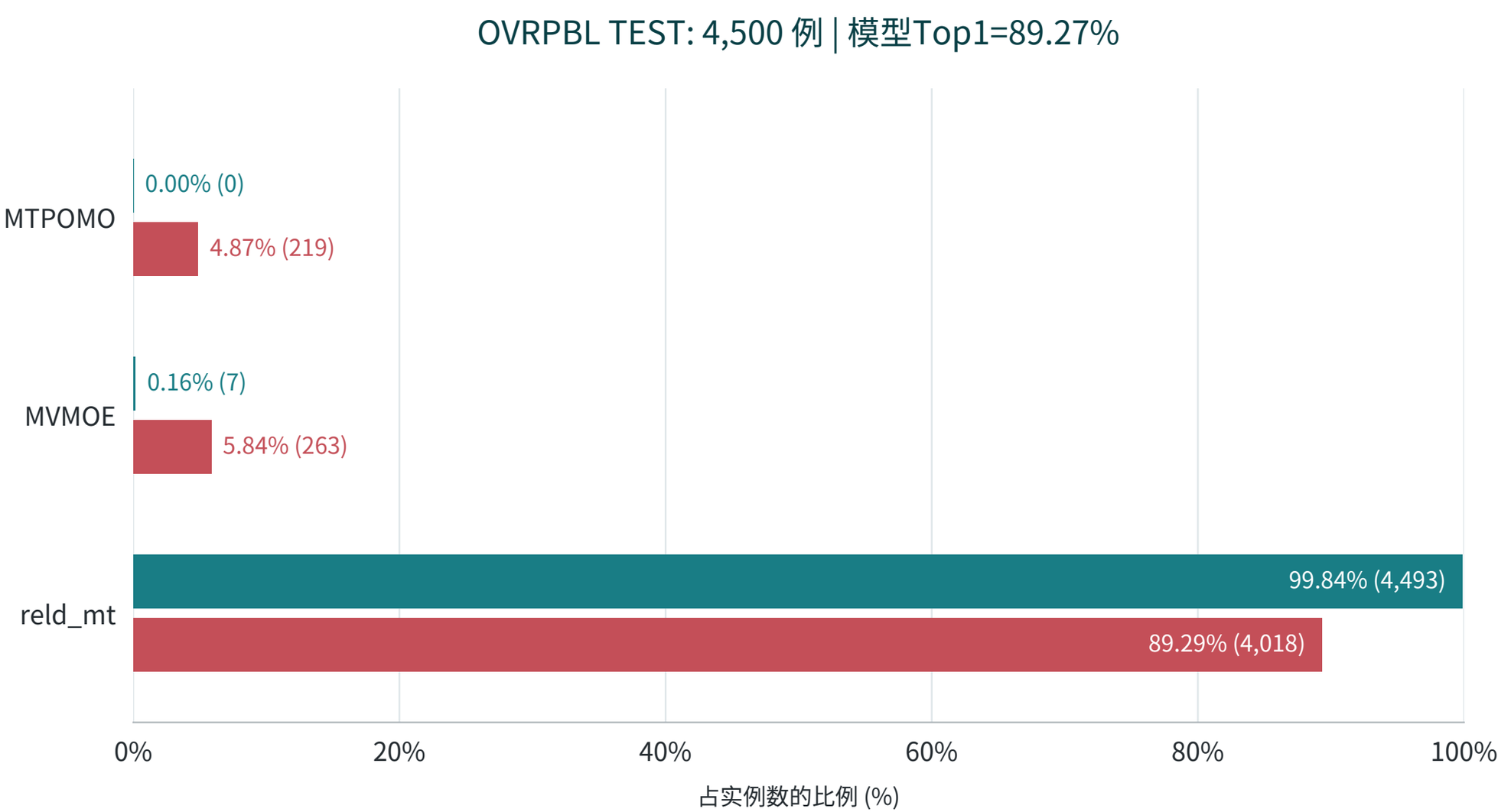
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

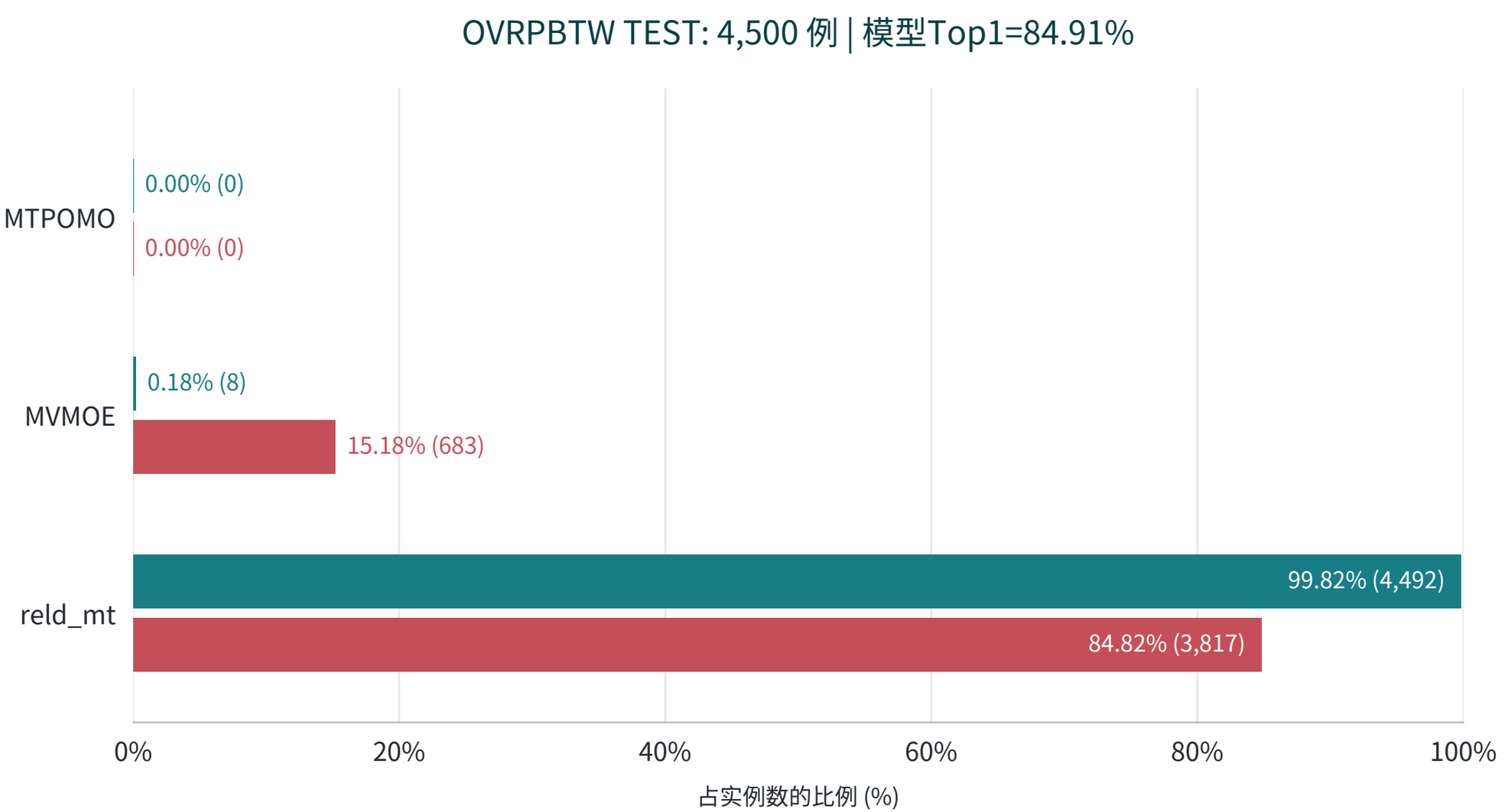
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

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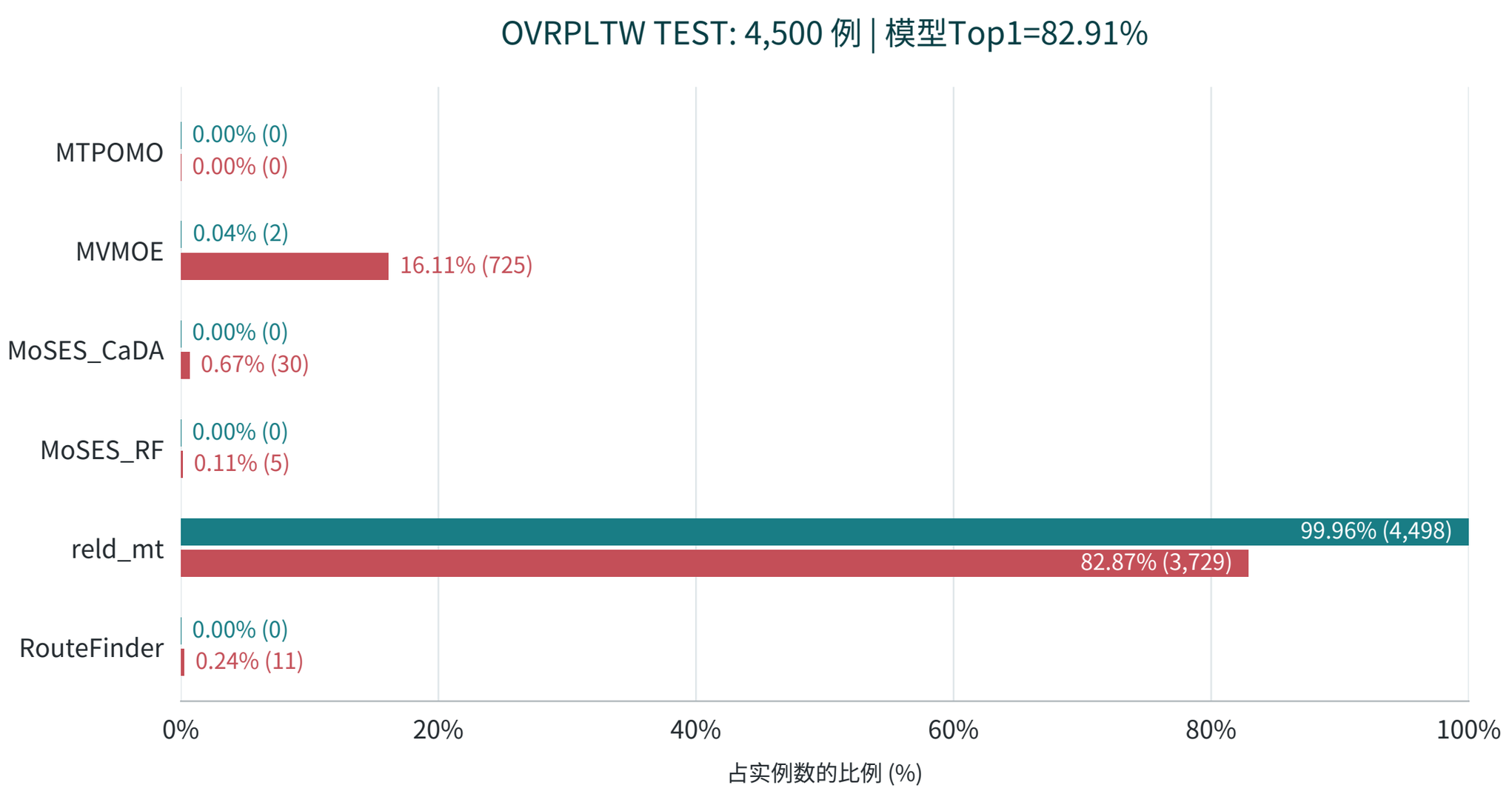
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | OVRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 5361f35200ded36a

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




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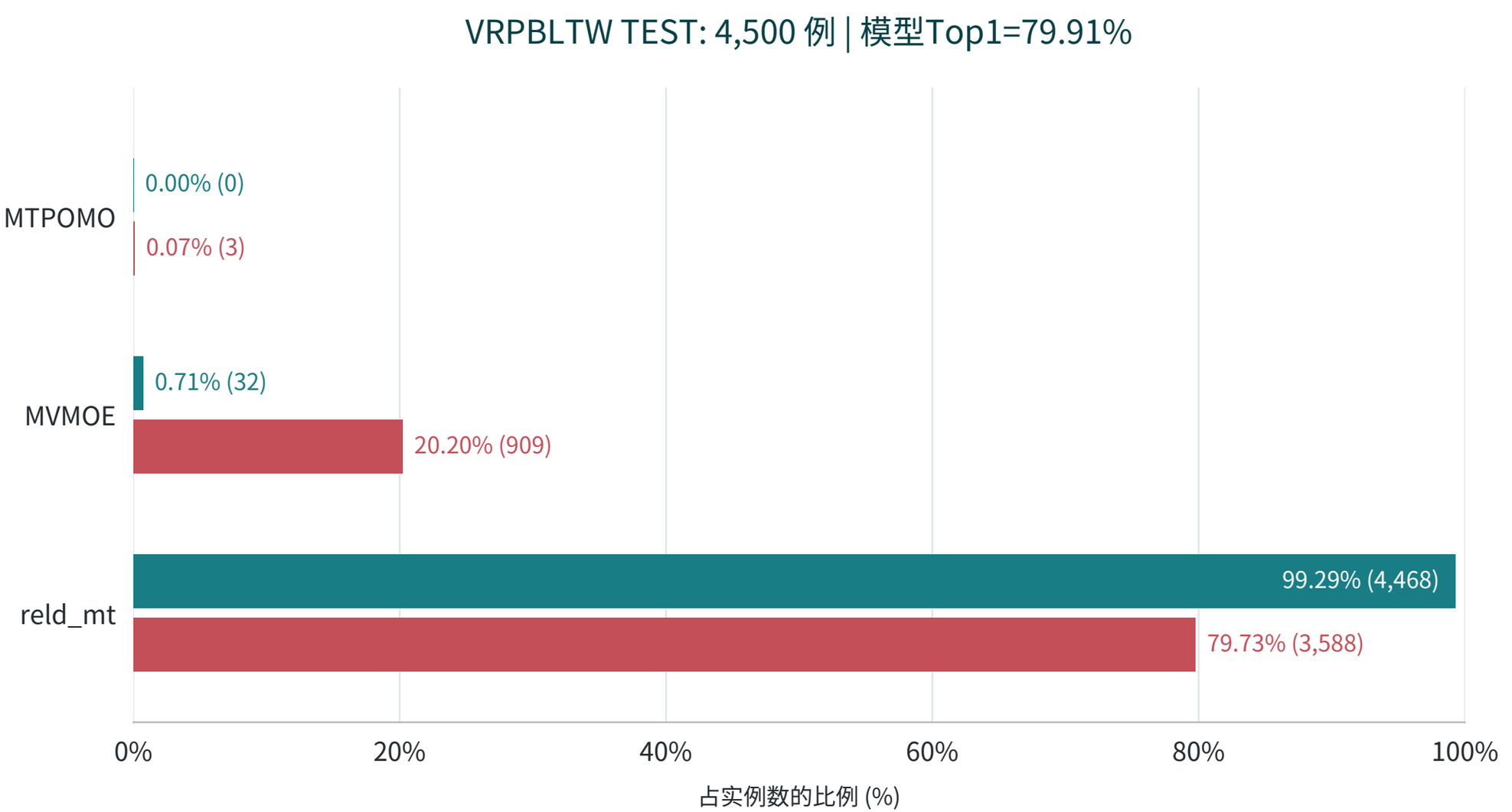
R85/B_gradient_balance: stable global solver-ID score ties. Same R85 lock; LIB inference only; no training/reselection/Rxx mixing.

# R85/B_gradient_balance | VRPBLTW | 方法分布

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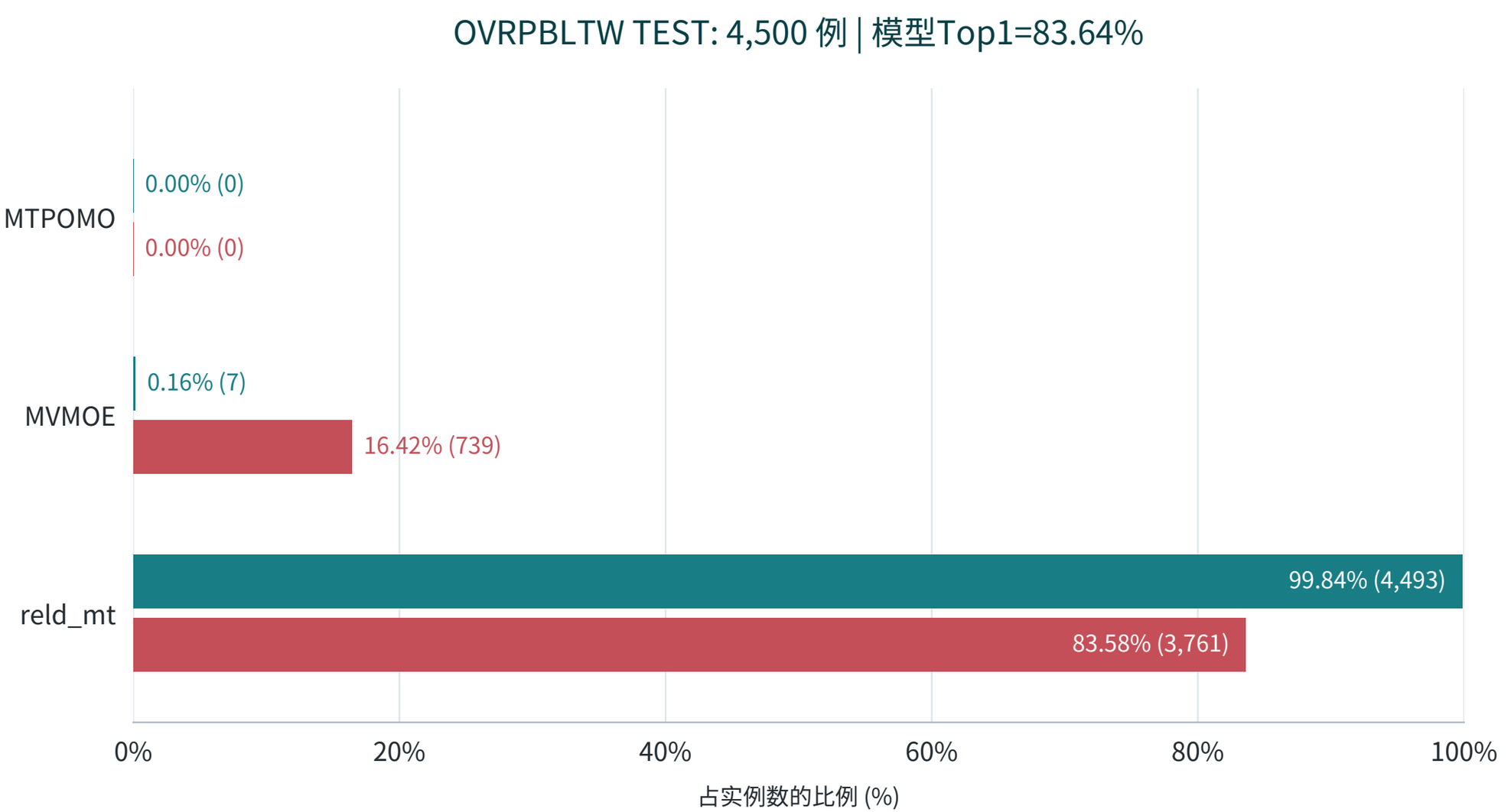
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